ZOUHEIR HAJJ SLEIMAN

Senior Structural Engineer | Structural Analysis & Design | M.E. (Structural), AUB

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PROFESSIONAL SUMMARY

Senior Structural Engineer with a Master of Engineering in Structural Engineering from the American University of Beirut and over a decade of hands-on experience in structural analysis, design, and site supervision. Graduate coursework concentrated in earthquake engineering, structural dynamics, and advanced analysis — applied directly to independent design practice covering reinforced-concrete residential and mixed-use buildings up to fifteen floors with multiple basement levels. Proficient in ETABS, CSI SAFE, SAP2000, Robot, Revit and AutoCAD; well-versed in ACI 318-19, UBC 97 and ASCE 7 (seismic and wind). Experienced in the full Lebanese permit process including OAR structural submissions and municipality approvals. A distinguishing aspect of this profile is the combination of structural design capability with direct construction supervision experience — providing practical understanding of constructability, site coordination, and the gap between drawings and execution that most design-office engineers rarely develop.

CORE COMPETENCIES

Structural Analysis Static Analysis • Response Spectrum Analysis • Seismic Analysis • Wind Load Analysis (ASCE 7) • Lateral System Design

Structural Design Reinforced Concrete Design (ACI 318-19) • RC Frame, slabs& Shear Wall Systems • Foundation Design • Retaining Walls • Transfer Slabs & Beams

Software ETABS • CSI SAFE • SAP2000 • Robot Structural Analysis • AutoCAD • Revit • SketchUp • MS Office Suite

Codes & Standards ACI 318-19 • UBC 97 • ASCE 7 • Lebanese NBC • IBC (familiarity)

Design Documentation Structural Calculation Reports • BOQ Preparation • Permit Drawings (OAR / Municipality) • Technical Specifications

Site & Constructability Construction Supervision • Quality Control • Structural Inspection • Coordination with Contractors & MEP teams

Languages Arabic (Native) • English (Fluent) • French (Working Proficiency)

PROFESSIONAL EXPERIENCE

Senior Structural Engineer & Engineering Manager *July 2021 – Present*

THE CUBE Construction | *Beirut, Lebanon*

- Perform independent structural analysis and design for concurrent residential, commercial, and institutional RC buildings using ETABS and CSI SAFE; develop and verify structural models, define load combinations, and produce design outputs in full compliance with ACI 318-19 and UBC 97 seismic requirements.
- Prepare complete structural calculation packages, reinforcement detailing drawings, and permit-ready documentation for OAR and municipality submissions under the Lebanese NBC framework.
- Design foundation systems, retaining walls, and basement structures for projects with varying site and soil conditions; coordinate structural layout and opening requirements with architectural and MEP disciplines during design development.
- Maintain design intent through the construction phase by supervising structural works on site, resolving RFIs, verifying reinforcement placement and concrete quality, and identifying non-conformances before they become embedded defects.
- Direct and technically mentor a team of more than ten engineers across design and site functions, reviewing their structural outputs and ensuring consistency with project specifications.
- Led the technical assessment and structural design for multiple Solar Water Pumping System (UNICEF and ESFD-funded), including feasibility study, structural sizing, and technical specification development.
- Manage project schedules, design deliverable timelines, and coordination meetings across concurrent projects, ensuring structural design packages are issued on time for permit submission and construction mobilisation.

- Oversee procurement of structural materials on selected projects, reviewing technical submittals, material certifications, and shop drawings for structural compliance prior to approval.
- Supervise structural works on site across multiple concurrent projects, conducting regular inspections of reinforcement, formwork, and concrete placement to ensure full compliance with structural drawings and project specifications.
- Coordinate and oversee subcontractor teams executing structural works, setting clear technical expectations, reviewing method statements, and managing performance to maintain quality and schedule on site.

Structural Design Engineer (Independent Practice) *May 2016 – July 2021*
Zouheir Hajj Sleiman Engineering Office | *Bekaa, Lebanon*

- Operated an independent structural engineering practice serving residential, commercial, and public-sector clients across the Bekaa region, providing full structural design and site supervision services.
- Designed reinforced-concrete structures for numerous residential and mixed-use buildings ranging from single-storey to G+12 with up to three basement levels, covering RC frame, flat slab, and shear wall systems.
- Performed structural modelling and analysis using ETABS and CSI SAFE; produced full structural calculation packages, reinforcement drawings, and permit-ready documentation for OAR and municipality submissions.
- Designed foundation systems including isolated footings, combined footings, and raft foundations calibrated to site soil conditions and loading requirements.
- Designed retaining walls and basement structures for sloped and constrained sites typical of the Bekaa region.
- Applied UBC 97 seismic provisions and ACI 318 design methodology consistently across all projects; coordinated with architectural and MEP teams during design development.
- Supervised construction of designed buildings on site, verifying compliance with structural drawings, reinforcement placement, concrete pouring sequences, and material specifications — providing direct constructability feedback into subsequent design iterations.

Junior Structural Engineer (Part-Time, concurrent with BE studies) *January 2012 – May 2016*
Hands Group | *Beirut, Lebanon*

- Assisted senior structural engineers in the analysis and design of residential and commercial buildings during bachelor's degree studies.
- Gained foundational experience in reinforced-concrete structural analysis, construction detailing, site inspection, and compliance with Lebanese code requirements.

EDUCATION

Master of Engineering (M.E.) — Structural Engineering *2017*
American University of Beirut (AUB) | *Beirut, Lebanon*

Key coursework: Earthquake Engineering & Structural Dynamics • Advanced Structural Analysis • Reinforced Concrete Design • Foundation Engineering • Finite Element Methods • Structural Reliability

Bachelor of Engineering (B.E.) — Civil & Environmental Engineering *2016*
Beirut Arab University (BAU) | *Beirut, Lebanon*

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- ETABS Structural Analysis & Design — Beirut Arab University
- ETABS Masterclass — Skill Booster
- CSI SAFE — Beirut Arab University
- Robot Structural Analysis (Steel Structures) — Udemy
- Project Management & Contract Administration — Order of Engineers and Architects, Beirut
- Execution Management — Order of Engineers and Architects, Beirut
- Class A Contractor Classification — Lebanese Syndicate of Public Contractors & Ministry of Public Works and Transportation